Machine Learning vs Deep Learning

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Course: Machine Learning Fundamentals

Workshop: 2.3 Assignment – Machine Learning vs Deep Learning

Overview

Machine Learning (ML) and Deep Learning (DL) are two important branches of artificial intelligence that help organizations solve business problems by learning from data and making predictions. Although both technologies have similar goals, they are designed for different types of applications. This report compares Gmail spam detection as a machine learning example and image recognition for employee check-in systems as a deep learning example. The report demonstrates how these technologies are used in real-world applications and explains why each approach is the most appropriate solution for its respective problem.

Machine Learning Example: Gmail Spam Detection

Email spam detection is one of the most common applications of machine learning. Gmail analyzes incoming emails using supervised learning algorithms that are trained with labeled examples of spam and legitimate emails. The model evaluates information such as keywords, sender details, links, and previous spam patterns to determine whether an email should be delivered to the inbox or moved to the spam folder. As additional emails are processed, the model continues to improve its accuracy and provides users with a safer and more reliable email experience.

Why Deep Learning Is Less Suitable: Traditional machine learning is the better choice for Gmail spam detection because the problem can be solved effectively using structured and labeled data. Deep learning would require much larger datasets, greater computing resources, and longer training time while providing little additional benefit for this type of classification problem.

Deep Learning Example: Image Recognition

Image recognition is a common application of deep learning that enables computer systems to identify people or objects directly from digital images. A practical example is an employee facial recognition system, where employees scan their faces to verify their identity before entering the workplace. Deep learning models such as Convolutional Neural Networks (CNNs) automatically learn facial features from images and compare them with stored data to improve workplace security and automate attendance tracking.

Why Machine Learning Is Less Suitable: Traditional machine learning relies on manually selecting image features before training the model. Because images contain complex visual patterns, this approach is less accurate and more time-consuming. Deep learning automatically learns these features from large datasets, making it a more effective solution for image recognition.

Conclusion

Machine learning and deep learning each provide valuable solutions for different business problems. Machine learning is well suited for structured data and classification tasks such as Gmail spam detection, while deep learning is more effective for complex applications involving images, speech, and other unstructured data. Selecting the appropriate approach depends on the nature of the problem, the available data, and the level of accuracy required. Understanding these differences enables organizations to implement AI solutions that best support their business objectives.

References

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